

Q1 2021 (31 Aug Shift 2)

Four identical hollow cylindrical columns of mild steel support a big structure of mass 50×10^3 kg. The inner and outer radii of each column are 50 cm and 100 cm respectively. Assuming uniform local distribution, calculate the compression strain of each column. [Use $Y = 2.0 \times 10^{11}$ Pa, $g = 9.8$ m/s²]

(1) 3.60×10^{-8}

(2) 2.60×10^{-7}

(3) 1.87×10^{-3}

(4) 7.07×10^{-4}

Q2 2021 (31 Aug Shift 1)

When a rubber ball is taken to a depth of m in deep sea, its volume decreases by 0.5%.

(The bulk modulus of rubber = 9.8×10^8 Nm⁻² Density of sea water = 10^3 kgm⁻³ $g = 9.8$ m/s²)

Q3 2021 (31 Aug Shift 1)

A uniform heavy rod of weight 10 kg ms², crosssectional area 100 cm² and length 20 cm is hanging from a fixed support. Young modulus of the material of the rod is 2×10^{11} Nm⁻². Neglecting the lateral contraction, find the elongation of rod due to its own weight.

(1) 2×10^{-9} m

(2) 5×10^8 m

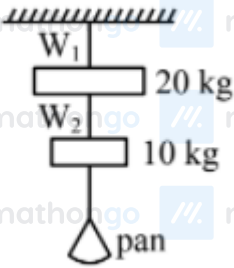
(3) 4×10^{-8} m

(4) 5×10^{-10} m

Q4 2021 (27 Aug Shift 2)

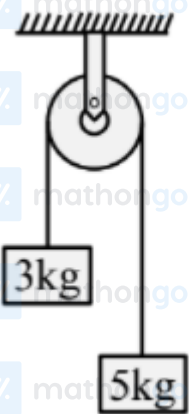
Wires W_1 and W_2 are made of same material having the breaking stress of 1.25×10^9 N/m² W_1 and W_2 have cross-sectional area of 8×10^{-7} m² and 4×10^{-7} m², respectively. Masses of 20 kg and 10 kg hang from them as shown in the figure. The maximum mass that can be placed in the pan without breaking the

wires is kg. (Use $g = 10 \text{ m/s}^2$)



Q5 2021 (26 Aug Shift 2)

Two blocks of masses 3 kg and 5 kg are connected by a metal wire going over a smooth pulley. The breaking stress of the metal is $\frac{24}{\pi} \times 10^2 \text{ Nm}^{-2}$. What is the minimum radius of the wire? (Take $g = 10 \text{ ms}^{-2}$)



(1) 125 cm

(2) 1250 cm

(3) 12.5 cm

(4) 1.25 cm

Answer Key

Q1 (2)

Q2 (500)

Q3 (4)

Q4 (40)

Q5 (3)

Q1 (2)

$$\text{Force on each column} = \frac{mg}{4}$$

$$\text{Strain} = \frac{mg}{4AY}$$

$$= \frac{50 \times 10^3 \times 9.8}{4 \times \pi(1 - 0.25) \times 2 \times 10^{11}}$$

$$= 2.6 \times 10^{-7}$$

Q2 (500)

$$B = -\frac{\Delta P}{\left(\frac{\Delta V}{V}\right)} = -\frac{\rho gh}{\left(\frac{\Delta V}{V}\right)}$$

$$-\frac{B \frac{\Delta V}{V}}{\rho g} = h$$

$$\frac{9.8 \times 10^8 \times 0.5}{100 \times 10^3 \times 9.8} = h$$

$$h = 500$$

Q3 (4)



We know,

$$\Delta \ell = \frac{WL}{2AY}$$

$$\Delta \ell = \frac{10 \times 1}{2 \times 5} \times 100 \times 10^{-4} \times 2 \times 10^{11}$$

$$\Delta \ell = \frac{1}{2} \times 10^{-9} = 5 \times 10^{-10} \text{ m}$$

Option (4)

Q4 (40)

$$B \cdot S_1 = \frac{T_{\max}}{8 \times 10^{-7}} \Rightarrow T_{1\max} = 8 \times 1.25 \times 100$$

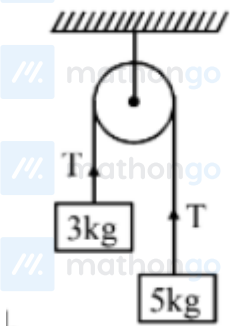
$$= 1000 \text{ N}$$

$$B \cdot S_2 = \frac{T_{2\max}}{4 \times 10^{-7}} \Rightarrow T_{2\max} = 4 \times 1.25 \times 100$$

$$= 500 \text{ N}$$

$$m = \frac{500 - 100}{10} = 40 \text{ kg}$$

Q5 (3)



$$T = \frac{2 m_1 m_2 g}{m_1 + m_2} = \frac{2 \times 3 \times 5 \times 10}{8}$$

$$= \frac{75}{2}$$

$$\text{Stress} = \frac{T}{A}$$

$$\frac{24}{\pi} \times 10^2 = \frac{75}{2 \times \pi R^2}$$

$$R^2 = \frac{75}{2 \times 24 \times 100} = \frac{3}{8 \times 24}$$

$$\Rightarrow R = 0.125 \text{ m}$$

$$R = 12.5 \text{ cm}$$